Appendix: Comparative Tool Matrix I

Symbol key (applies to all cells): **Y** = full native support · ∼ = partial or plugin-based · **N** = absent. See also Figure 4 for a colour-coded heatmap rendering of this table.

Table 1: Comparative feature matrix (14 capabilities × 10 tools). Y = full native support, ∼ = partial or plugin-based, N = absent.

Capability	`template/`	Snakemake 9	Nextflow 25	CWL 1.2	Quarto 1	Jupyter Book 2	R Markdown	DVC 3	Overleaf (2025)	OpenAI Prism
Pipeline orchestration	Y	Y	Y	Y	∼	N	N	Y	N	N
Manuscript rendering	Y	N	N	N	Y	Y	Y	N	Y	Y
Testing enforcement	Y	N	N	N	N	N	N	N	N	N
Coverage thresholds	Y	N	N	N	N	N	N	N	N	N
Cryptographic provenance	Y	N	∼[1]	N	N	N	N	∼[2]	N	N
Steganographic watermarking	Y	N	N	N	N	N	N	N	N	N
Multi-project management	Y	N	N	N	N	N	N	N	∼	∼
AI-agent documentation	Y	N	N	N	N	N	N	N	∼	∼

Appendix: Comparative Tool Matrix II

Agentic skill protocol (SKILL.md / MCP)	Y	N	N	N	N	N	N	N	N	N
Interactive TUI	Y	N	N	N	N	N	N	N	N	N
Zero-mock policy	Y	N	N	N	N	N	N	N	N	N
Container support	N	Y	Y	Y	N	N	N	N	N	N
Distributed execution	N	Y	Y	Y	N	N	N	~[3]	N	N
Multi-language (R/Julia)	N	Y	N	Y	Y	Y	Y	Y	N	N

Appendix: Comparative Tool Matrix III

[1] Nextflow 25.04.0 introduced data-lineage provenance tracking (build-level, not document-level). [2] DVC provides content-addressed versioning for data artifacts via its object store. [3] DVC integrates with remote storage (S3, GCS, Azure) but does not natively orchestrate distributed compute. Overleaf and OpenAI Prism are collaborative cloud LaTeX/AI writing environments; their AI features (GPT-5.2 for Prism, Overleaf Labs AI for Overleaf) are partial/early-stage as of 2025–2026.